

JECRC University, Jaipur
B.Tech.-I Sem I Year 2025-26
Tutorial Sheet-11 (CO4)
Subject: Mathematics-1
Code: DMA024A

CO4: To deal with functions of several variables that are essential in most branches of engineering.

Part-A (2x5=10)

A1. Define Homogeneous function.

A2. State the Euler's theorem for homogeneous function of degree n.

A3. If $z = x^2y^2 + xy^3 + y^4$ the find $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}$.

A4. Find the equation of the tangent to the curve $f(x) = x^3 - 3x^2 + x - 1$ at the point where $x = 3$.

A5. Find the slop of tangent line of the curve $y = x^2 - 5x + 6$ at the point (2, 0).

Part –B

(3x7=21)

B1. If $u = \sec^{-1} \left(\frac{x^3+y^3}{x+y} \right)$, then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u$.

B2. Verify Euler's theorem for the function $z = \frac{x^2+y^2}{x+y}$.

B3. Find the equation of tangent and normal to the curve $y(x-2)(x-3) - x + 7 = 0$ at the point where it cut the x-axis.

Part –C

(3x11=33)

C1. If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, prove that $xu_x + yu_y + zu_z = 0$.

C2. If $U = x \sin^{-1} \frac{y}{x}$ then prove that $x^2 \frac{\partial^2 U}{\partial x^2} + 2xy \frac{\partial^2 U}{\partial x \partial y} + y^2 \frac{\partial^2 U}{\partial y^2} = 0$

C.3. Find the equation of tangent and normal to the curve $x = \frac{2at^2}{1+t^2}$, $y = \frac{2at^3}{1+t^2}$ at the point $t = \frac{1}{2}$.

Answers:

A1.	A2.	A3.	A4. $y = 10x - 28$	A5. 1	B1.
B2.		B3. $x - 20y - 7 = 0$ $20x + y - 140 = 0$	C1.	C2.	C3. $13x - 16y - 2a = 0$ $16x + 13y - 9a = 0$